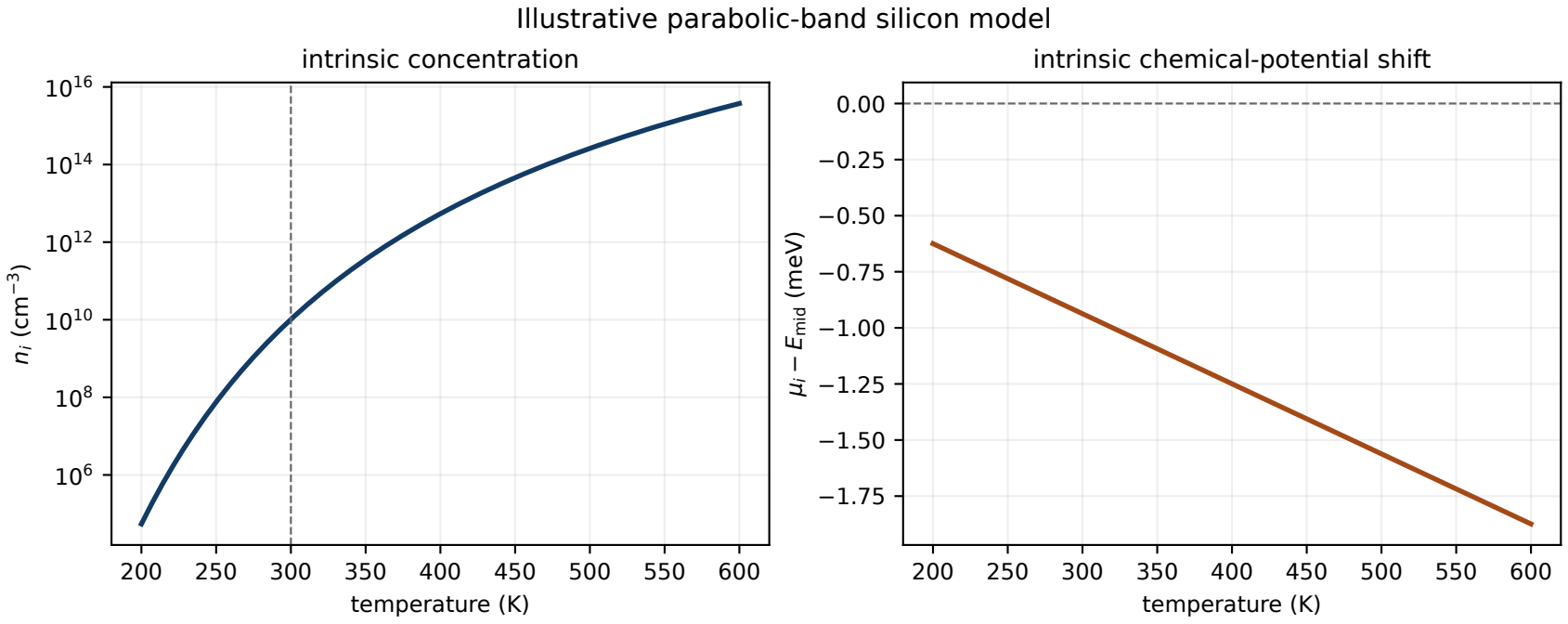
Illustrative parabolic-band silicon model
intrinsic concentration
n_i (cm^{-3})
10^{16}
10^{14}
10^{12}
10^{10}
10^{8}
10^{6}
200
250
300
350
400
450
500
550
600
temperature (K)
intrinsic chemical-potential shift
$\mu_i - E_{\mathrm{mid}}$ (meV)
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
200
250
300
350
400
450
500
550
600
temperature (K)